Viraj Nalbalwar

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Education

COEP Technological University **July 2024 - May 2027**
B.Tech - Computer Science and Engineering - Final Year ***(CGPA: 8.74)*** *Pune, Maharashtra*

Dr. Babasaheb Ambedkar Technological University **Sep 2021 - May 2024**
Diploma in Computer Engineering ***(CGPA: 9.92, State Rank: 2)*** *Raigad, Maharashtra*

Experience

Software Engineer **May 2026 - July 2026**
Mastercard Technology Private Limited *Pune, Maharashtra*

- Developed Apache Spark-based Java Spring Boot migration services to transfer enterprise financial data from legacy DB2 systems to Oracle databases.
- Built REST-triggered batch migration workflows with Apache Spark JDBC reads, partition-based processing, audit metadata handling, and Oracle target writes.
- Added thread-safe execution controls for migration endpoints to prevent concurrent job conflicts during long-running data loads.

Cloud Engineer **June 2025 - July 2025**
Reliance Jio Platforms Ltd. *Navi Mumbai, Maharashtra*

- Deployed and managed OpenStack environment on RHEL, configuring core services (Keystone, Nova, Neutron, Glance, Cinder) to enable private cloud compute and storage orchestration.
- Configured Docker and Kubernetes workloads and built monitoring using Prometheus and Grafana to track system health, network performance, and application metrics.

Responsive Web Design Intern **June 2023 - July 2023**
SEED Infotech Ltd. *Pune, Maharashtra*

- Developed production-ready web interfaces using Next.js, React.js, and Tailwind CSS, implementing component-based architecture and optimized routing for performance.
- Followed modern development practices using Git, code reviews, and Lighthouse audits to ensure fast sub-second page load times.

Projects

Thermal-Aware Core-Level Scheduling for Kubernetes Workloads **Dec 2025 - Feb 2026**
Kubernetes, Go, Linux, DaemonSets

- Built a thermal-aware Kubernetes scheduling system that integrates per-core temperature and utilization telemetry to improve pod placement and reduce thermal throttling.
- Developed Linux cgroups v2-based CPU pinning mechanisms to enable intelligent core-level workload allocation.

CloudInfraLab - OpenStack Deployment **June 2025 - July 2025**
RHEL, XRDP, DevStack, Docker, Kubernetes

- Architected full-scale cloud lab with manual OpenStack configuration on RHEL 9.2, integrating core services for identity, image management, compute, networking, and block storage.
- Automated virtual machine provisioning and containerized workloads using DevStack to simulate production-grade private cloud environments.

SteganoSafe - Invisible Ink for the Digital Age **Dec 2024 - Jan 2025**
Next.js, TypeScript, Tailwind CSS

- Developed web application implementing LSB steganography for secure message embedding within images, processing data entirely in-browser for enhanced privacy.

Technical Skills

Java, TypeScript, SQL, Next.js, Kubernetes, Docker, OpenStack, OpenShift, Linux, Prometheus, Grafana, GitHub, iLO

Recognitions

Energy Efficiency with Sustainable Performance: Techniques, Tools, and Best Practices **June 2026**
Organized by: ISC High Performance, Hamburg, Germany

- Accepted research paper: **"Thermal-Aware Core-Level Scheduling for Kubernetes Workloads"**

Gold Medalist - Diploma in Computer Engineering **May 2025**
Dr. Babasaheb Ambedkar Technological University, Lonere